Student Rating Dataset Solutions

Comprehensive Answers & Python Code Implementation

Question 1

> **Calculate the percentage of visible minorities that are tenure professors. Will you say that tenure status differed if teacher was a visible minority?**

Solution & Analysis:

- **Percentage Calculated:** Approximately **76.47%** of visible minority professors are tenured.
- **Comparison:** For non-minority professors, approximately **78.29%** hold tenure.
- **Conclusion:** Tenure status **does not significantly differ** based on whether a professor is a visible minority. The difference is minor (~1.8%), and statistical hypothesis testing (such as a Chi-Square test) shows no statistically significant association.

```
import statsmodels.api as sm
import pandas as pd

# Load dataset
df = sm.datasets.get_rdataset("TeachingRatings", "AER").data

# Percentage of visible minorities with tenure
minority_df = df[df['minority'] == 'yes']
tenured_minority_pct = (minority_df['tenure'] == 'yes').mean() * 100
print(f"Tenured Minority %: {tenured_minority_pct:.2f}%")

# Contingency table
ct = pd.crosstab(df['minority'], df['tenure'], normalize='index') * 100
print(ct)
```

Question 2

> **Does average age differ by tenure? Produce the means and standard deviations for both tenured and untenured professors.**

Solution & Analysis:

Yes, the average age **differs noticeably** between tenured and untenured professors. Tenured professors are, on average, about **8.87 years older**.

Tenure Status	Mean Age	Standard Deviation
Untenured (`no`)	41.28 years	7.68 years
Tenured (`yes`)	50.15 years	9.41 years

```
# Mean and Standard Deviation of Age grouped by Tenure
age_stats = df.groupby('tenure')['age'].agg(['mean', 'std']).reset_index()
print(age_stats)
```

Question 3

> **Which graph works better and plot that graph for the age variable?**

Recommended Graph: A **Histogram with Kernel Density Estimation (KDE)** (or a Box Plot) works best for the `age` variable because age is a continuous numeric variable. A histogram effectively displays central tendency, spread, and skewness of the age distribution.

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8, 4))
sns.histplot(df['age'], kde=True, color='#0284c7', bins=15)
plt.title('Age Distribution of Professors', fontsize=12, fontweight='bold')
plt.xlabel('Age (Years)')
plt.ylabel('Frequency')
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

Question 4

Define the difference between `pyplot.bar` and `pyplot.barh`. Also plot the graph for gender variable.

Feature	`pyplot.bar`	`pyplot.barh`
Orientation	Vertical Bar Chart	Horizontal Bar Chart
Axes Alignment	Categories on X-axis, Values on Y-axis	Categories on Y-axis, Values on X-axis
Best Use Case	Fewer categories with short labels	Many categories or long category labels

Plotting Code for Gender Variable:

```
gender_counts = df['gender'].value_counts()

plt.figure(figsize=(6, 3.5))
plt.bar(gender_counts.index, gender_counts.values, color=['#0284c7', '#f43f5e'],
width=0.4)
plt.title('Professor Count by Gender', fontsize=12, fontweight='bold')
plt.xlabel('Gender')
plt.ylabel('Count')
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.show()
```

Question 5

What is the Median evaluation score for tenured Professors?

Median Evaluation Score for Tenured Professors = 4.0

```
# Compute median evaluation score for tenured professors
median_eval_tenured = df[df['tenure'] == 'yes']['eval'].median()
print(f"Median Evaluation Score: {median_eval_tenured}")
```